DELFT UNIVERSITY OF TECHNOLOGY
Faculty of Electrical Engineering, Mathematics and Computer Science

# CSE2315 Lab Course
# Assignment 6

- This assignment consists of several exam-level questions and sometimes an optional extra exercise.
- See Brightspace, *Course Information* for instructions about the lab course, rules and grading procedure.
- Your solution has to be handed in on paper, typeset in LaTeX, using a word processor or readable handwriting. Handing in via email is not permitted.
- Cooperating with a colleague is permitted and *strongly encouraged!* In such cases, please hand in a single copy of the work.
- Total number of pages, without this cover page: 2.

1. Suppose we have the following language:

$$E_{\mathsf{GNFA}} = \{\langle G\rangle \mid G \text{ is a GNFA such that } L(G) = \varnothing\}.$$

Show that $E_{\mathsf{GNFA}}$ is decidable. You are allowed to make use of the deciders implied by the text in the book (e.g., Section 4.1).

2. Consider the language

$$L = \{\langle M, w\rangle \mid M \text{ halts on } w \text{ and moves the head at most } |w|^2 \text{ steps}\}.$$

Is this language decidable? Explain.

3. *Old exam question, endterm 20-21:* Show the claim is true or give a counterexample and explanation to show the claim is false, starting your answer with either *True* or *False*.

If there is a Turing reduction from $L$ to $E_{CFG}$, then $\overline{L}$ is Turing-recognisable.

4. *Old exam question, endterm 22-23:*

Consider the following language:

$$L = \{\langle M, q\rangle \mid M \text{ is a Turing Machine and } q \text{ is a useless state in } M\}$$

A useless state in a Turing Machine is one that is never entered on any input string.

Prove $L$ is undecidable, using a Turing reduction $E_{\mathsf{TM}} \leq_T L$.

5. *Old exam question, resit 17-18:* Suppose we have the following language

$$L = \{\langle M, v, G\rangle \mid M \text{ outputs } G \text{ on the tape, which is a context free grammar that accepts } v \text{, when run on input } v.\}$$

We intend to prove that this language is undecidable. To that end we use a mapping reduction from $HALT_{TM}$ to $L$. The reduction $f$ is computed by the following TM $F$:

$F =$"On input $\langle M, w\rangle$:

   1. Construct the following TM $M'$:
      $M' =$"On input $x$:
      1. Clear the input from the tape.
      2. Run $M$ on $w$.
      3. See question a.
      4. See question a.
   2. Write $\langle M', a, G\rangle$ to the output tape, where $G$ is a grammar with one rule: $S \to a$ and starting variable $S$."

   (a) What should be done in steps 3 and 4 of $M'$?

   (b) Provide a proof showing $f$ satisfies the requirements of a mapping reduction.

6. *Old exam question, resit 20-21:*

   (a) Let $MIA_{TM}$ be the language consisting of 3 TM's and a word such that the word is recognized by at least 2 of the 3 TM's, formally described as $MIA_{TM} = \{\langle M_1, M_2, M_3, w\rangle \mid M_1, M_2 \text{ and } M_3$ are TM's and $w$ is a word such that $w \in (L(M_1) \cap L(M_2)) \cup (L(M_1) \cap L(M_3)) \cup (L(M_2) \cap L(M_3))\}$. Show that this is a recognizable language.

   (b) Show that $MIA_{TM}$ is undecidable by constructing a Turing reduction from $A_{TM}$.

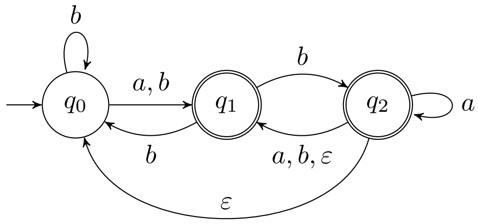
   (c) Show that $MIA_{TM}$ is not co-turing recognizable.

7. *Revision, old exam question, midterm 19-20:*

(a) Assume that we define an expanded NFA model by imposing the additional requirement that $|F| = 1$, and we call this an SNFA. Are the SNFA and NFA models equally expressive? Start your answer with "Yes" or "No". If you answer yes, explain how an arbitrary SNFA can be rewritten as an NFA and vice versa. If you answer no, give an example of a language that can be accepted by an NFA, but not by an SNFA or vice versa.

(b) Give an NFA of at most 4 states that accepts the word $mia$ and rejects the word $maya$. No other restrictions on the language apply. A transition diagram suffices.

(c) Consider the NFA $N$ depicted as:




Using the method from Sipser and leaving out unreachable states, construct a new DFA $D$, such that $L(D) = L(N)$. A transition diagram and a short explanation suffice.